



SEMESTER - III

BRANCH	SubjectTitle	SubjectCode
B.Sc.(Botany-Minor)	Industrial Botany	

Course Objective :-The Objective of this paper is to gain thorough knowledge about various plant groups from primitive to highly evolved plants. Students become aware of applications of different plants in various industries. To highlight the potential of these studies to become an entrepreneur.To equip the students with skills related to laboratory as well as industries based studies.To make the students aware about conservation and sustainable use of plants.To address the socio-economic challenges related to plant sciences.

Course Learning outcomes(CLO)

- This course will provide knowledge on plants and their parts used in various industries.
- Students will get an idea to establish plant based natural product industry.
- This course will make the students self-reliant.

Content

Unit -I

Plants in Timber Industry:

- 1.1 Timber yielding trees of India and their products (Shisham, Sal, Teak, Deodar, Babool).
- 1.2 Bamboo and Cane Industry.
- 1.3 Kattha' Industry.

Unit-II

Leaf Based Industries-

- 1.1 Utility products of leaf (Palash, Banana).
- 1.2 Tea Industry (Production of various types of teas).
- 1.3 Leaf oil Industry (Mint, Camphor, Neem, Tulsi, Eucalyptus and Lemon grass).
- 1.4 Leaves used as spices (Kasoori Methi, Pudina, Curry pattà. Onion, Tejpatta)

Unit-III

Flower based Industries-

- 1.1 Perfume products of Gulab, Jasmine, Henna,
- 1.2 Color industry (Food and Holi colors).
- 1.3 Raw material for Fermentation (Mahua).



Unit-IV

Fruits and Seedsbased Industries-

- 1.1** Jams, Jellies, Juice, Sauce and Pickles.
- 1.2** Poha and Daal Industry.
- 1.3** Edible Oil Industry (Groundnut, Soybean)
- 1.4** Starch, Glucose, and Dextrose Industry.

Unit-V

Other parts of plants based Industries-

- 1.1** Sugar and Jaggery Industries.
- 1.2** Jute and Agarbatti stick making industry.
- 1.3** Project proposal preparation for establishment of an industry.
- 1.4** Grants and funding provider organizations of India.

Practicals

- 1. Preparation of Holi color's from locally available flowers
- 2. Preparation of food colors from locally available flowers
- 3. Perfume extraction process by distillation method
- 4. Preparation and preservation techniques of jams, jellies and prickles.
- 5. Extraction and preservation of juices (lemon and orange etc.)
- 6. Preparation of different types of teas (Tulsi tea, lemon tea etc.)
- 7. Identification, collection and extraction of oil yielding leaves.
- 8. Identification, collection and specimen preparation of leafy spices.
- 9. Hands on training for preparation of "Douna and Pattal" using Palash and Banana leaves.
- 10. Visit to any plant based industry.
- 11. Herbarium preparation of different parts of plants used in various industries



RAMKRISHNADHARMARTHFOUNDATIONUNIVERSITY,BHOPAL

AirportBypassRoad,GandhiNagar,Bhopal-462033
